NETWORKING FUNDAMENTALS 101

Week 2 Practical — Network Topologies and Connections

Cisco Packet Tracer Lab

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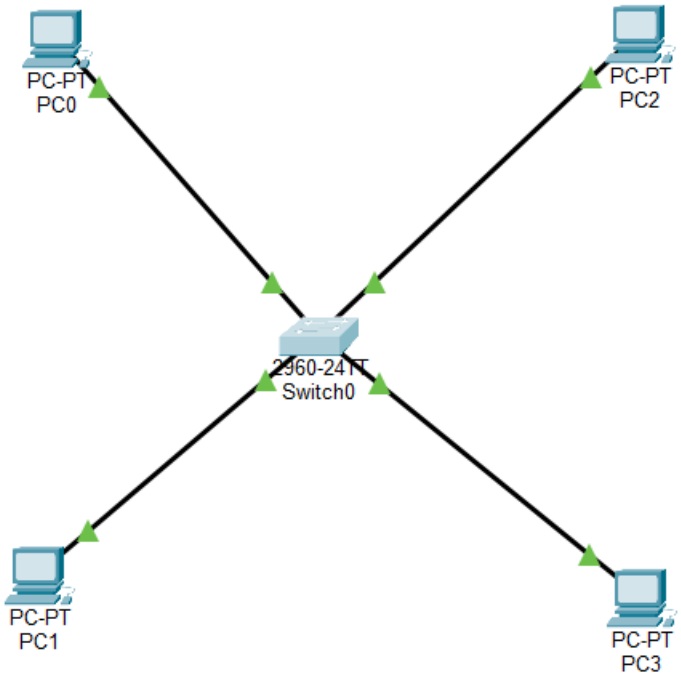
Date:	13–19 March 2025	**Total Marks:**	/ 5

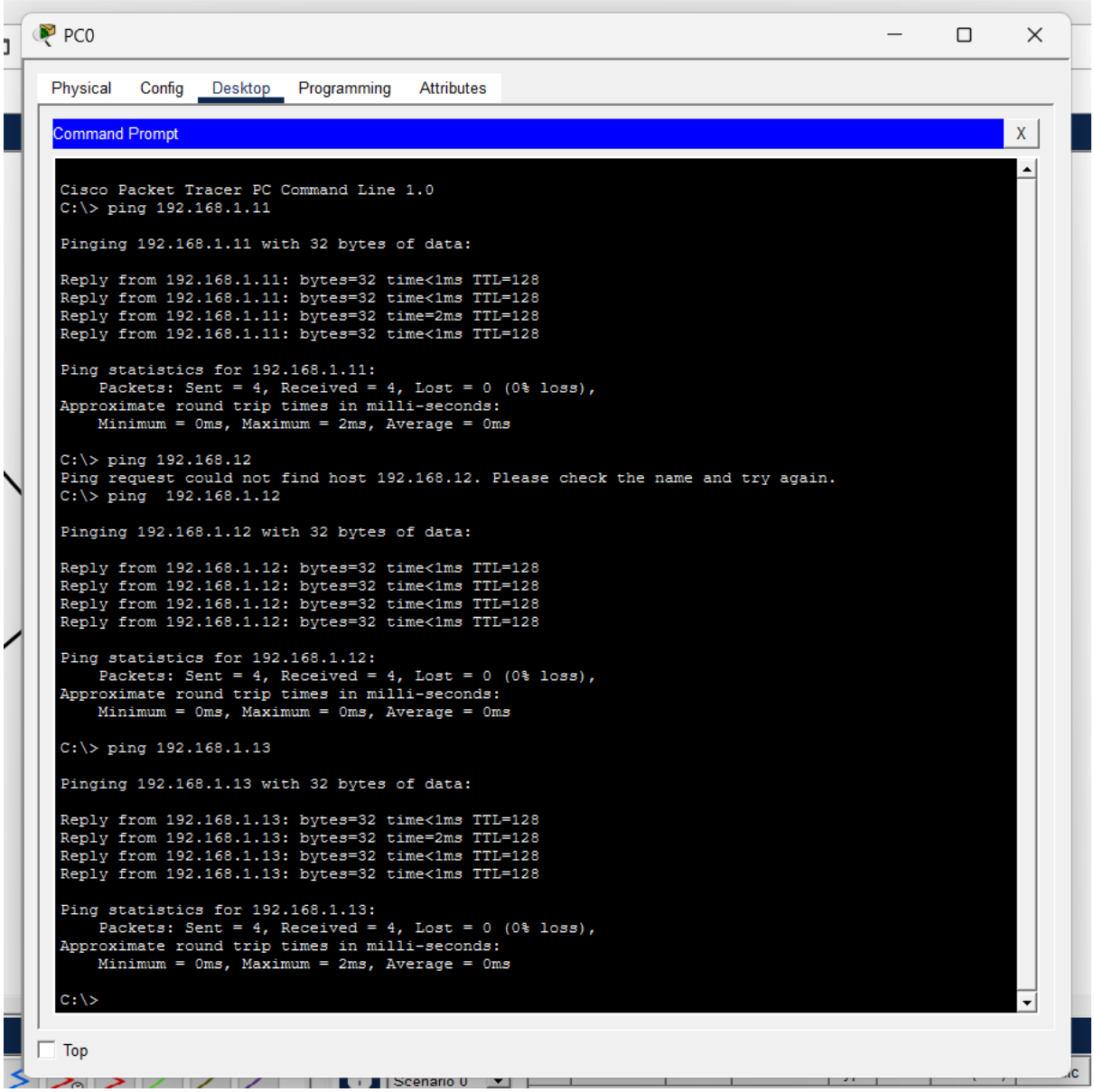
1. TASK1:STAR TOPOLGY



1.2.VERIFICATION OF LINK STATUS,DUPLEX,AND SPEED

Port	Connected Device	Status	Duplex	Speed
Fa0/1	**PC0**	connected	a-full	a-100
Fa0/2	**PC1**	connected	a-full	a-100
Fa0/3	**PC2**	connected	a-full	a-100
Fa0/4	**PC3**	connected	a-full	a-100

1.3.PING TEST


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\> ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=2ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 0ms
C:\> ping 192.168.12
Ping request could not find host 192.168.12. Please check the name and try again.
C:\> ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\> ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=2ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 0ms
C:\>
Top

Source	Destination	IP Address	Result (Success/Fail)
PC0	PC1	192.168.1.11	Success
	PC2	192.168.1.12	Success
	PC3	192.168.1.13	Success

1.4.

There is no communication between the PCs. Their means of connection to each other and
communication has been interrupted when the switch was removed.

Q2: Is a star topology fault tolerant? Justify your answer with reference to your observations.
NO.
All devices lost their connection to the network because the switch was removed.

Task 1 Marking Criteria	Available	Awarded
Star topology correctly built with all devices and IP addresses	**0.5**	
Link status/duplex/speed table completed from CLI output	**0.5**	
Ping test results recorded with supporting screenshot	**0.5**	
Fault tolerance questions answered with clear reasoning	**0.5**	
TASK 1 TOTAL	**2**	